MILITARY SPECIFICATION

TRAILER, SEMITRACTOR, END DUMP, 22 CUBIC YARD CAPACITY, DED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-propelled, pneumatic tired, diesel engine driven, hydraulic operated, 22 cubic yard (cu yd) minimum struck capacity, trailer, semitractor, end dump, (hereinafter referred to as the unit) with features as specified herein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-F-800 - Fuel Oil, Diesel.

MILITARY

MIL-V-173 - Varnish-Moisture-and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment.

MIL-T-3351 - Tractor, Full-Tracked, Low-Speed; Tractor Wheeled, Agricultural; and Tractor Wheeled, Industrial: and Their Attachments, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3805

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies of DoL should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096).

TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description. The unit shall be a self-propelled, diesel-engine-driven hydraulic operated, semitractor, two-driven wheels, trailer end dump, two-axle type, 22 cu yd minimum struck capacity.

3.2 First article. When specified (see 6.2), the contractor shall furnish a complete unit for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The unit shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Construction. The unit shall be designed and constructed to facilitate field maintenance. All adjustments and replaceable accessories shall be readily accessible. Conditions which can be hazardous to personnel or deleterious to equipment shall not be permitted.

3.5 Identical items. All units furnished under a specific contract shall be physically and mechanically identical. Parts, accessories, assemblies, and components are included in this requirement. Written approval for deviations must be obtained in advance from the contracting officer.

3.6 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.7 Safety. All moving parts, parts subject to high operating temperatures, and parts energized electrically which are of such a nature or so located as to become a hazard to operating and maintenance personnel shall be insulated, enclosed, or guarded to the extent necessary to eliminate the hazard. Protective devices shall not impair the operating functions. Platforms, steps, handholds, and rails shall be furnished and located where necessary, to assure safe, easy access by personnel to those areas requiring frequent maintenance or service. All walking surfaces shall be covered with antiskid material or painted with a nonskid abrasive type deck paint. The unit shall comply with OSHA regulations in effect at the time of manufacture. The unit shall be provided with the following equipment:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.

a. Safety glass windshield with power operated windshield wiper(s).
b. Horn.
c. Backup alarm device (conforming to SAE J994).
d. Seat belt for operator's seat (conforming to SAE J386).
e. Rollover protective structure (conforming to SAE J1040).
f. Adjustable air suspension seat.
g. Instrument panel (vandalism) protective guard.
h. Fenders over driving wheels.
i. Emergency steering system (conforming to SAE J53).

3.8 Dimensions, weights, and capacity. The unit shall conform to the dimensions, weights, and capacity specified in table I. Struck capacity yardage rating shall conform to SAE J741.

TABLE I. Dimensions, weights, and capacities.

Struck	22 cu yd minimum
Payload	66,000 lb minimum
Overall length	33 feet minimum
Wheel base (haul position)	21 feet minimum
Dumping angle	50° minimum
Nonstop turning width (body down)	34 feet maximum
Shipping weight	28,000 lb minimum
Travel speed mph (empty on level earth haul road)	24 mph minimum

3.9 Performance. The unit shall withstand the hard usage encountered, without permanent deformation or damage, while performing the following operations:

a. With a minimum struck payload as specified in table I, travel at an average speed of not less than 23 miles per hour (mph) on maintained earth haul roads (see 6.5.1).
b. With a minimum struck payload as specified in table I, negotiate a grade having a minimum slope of not less than 20 percent.
c. Without payload, traverse a longitudinal grade having a minimum slope of not less than 30 percent.
d. Haul and dump the load, without assistance, from auxiliary equipment.

3.10 Environmental conditions. The tractor engine shall be capable of starting and satisfactorily performing at a temperature from -20° Fahrenheit (F) to +120°F, and at any elevation from sea level up to a 5,000 foot elevation. When an ether priming system is required, it shall be of the measured shot type with storage capacity of not less than 12 fluid ounces (see 6.2).

3.11 Tractor frame. The tractor frame shall support or incorporate as integral parts the bumper, towing device(s), crankcase guard, fenders, oscillating hitch assembly, operator's station, powerplant, and power train. Frame shall be mounted on two power driven pneumatic-tired wheels. Towing device(s) shall withstand a pulling force equivalent to not less than 150 percent of the unit's gross vehicle weight (gvw) (see 6.5.2). Crankcase guard shall protect the crankcase, flywheel housing, accessories, and other assemblies exposed to damage from projecting obstructions likely to be

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.

encountered during normal operations. Crankcase guard shall be furnished with openings which shall provide easy access to points requiring periodic servicing and cleaning. Access openings shall be equipped with cover plates.

3.12 Trailer. The trailer shall have a minimum struck capacity of 22 cu yd when measured in accordance with SAE J741 for yardage ratings. The unit shall have a payload of not less than 66,000 pounds (lb). Trailer to be a single nondriven axle with two wheels which is rigidly attached to the trailer body.

3.13 Brakes.

3.13.1 Service brakes. The unit shall be equipped with full air or air over hydraulic service brakes on all wheels, using either a drum or disk type brake. The brakes shall be capable of stopping the fully loaded unit from a speed of 20 mph within 90 feet from the point of brake pedal application. Brakes shall be equipped with an automatic safety device to cause application of the brakes, or provide emergency braking, in the event of loss of air and to give an audible signal to the operator when the loss occurs. Linings for drum-type brakes shall be either bonded, riveted, or bolted to the brake shoes. Air reservoir(s) shall have moisture bleeder(s). The brake system shall be synchronized to apply the trailer brakes prior to application of the tractor brakes.

3.13.2 Parking brake. The unit shall be equipped with an independent parking brake system capable of holding the fully loaded unit on a 15 percent grade without the assistance of the service brakes. This independent parking brake system also serves to lock the trailer brakes while dumping.

3.14 Wheels, rims, and tires. The tires shall be of the same size and interchangeable front to rear. Tires shall be wide base, tubeless, traction tread (E-2). Tire and rim sizes, ply rating, and load rating at 30 mph shall conform to T&RA recommendations. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground with unit loaded to rated gvw (with overload not to exceed 6 percent) (see 6.5.2).

3.15 Power plant. The unit shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle, diesel engine. The net horsepower rating at the flywheel shall be not less than 330 when determined as specified in SAE J816. The engine shall be capable of operation on diesel fuel conforming to VV-F-800. The unit shall be furnished complete with the following accessories:

a. Two stage intake air cleaner with service indicator.
b. Full flow oil filter with bypass.
c. Primary and secondary fuel filters.
d. Hour meter.
e. Fuel tank with fuel level indicator and sufficient capacity for 8 hours operation.
f. Lubricating oil pressure indicator.
g. Coolant temperature indicator.
h. Electric cranking system.
i. Battery(s), dry charged, without a filling of electrolyte.
j. Battery charging alternator/rectifier system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.

k. Battery charging indicator.
l. Radiator guard.
m. A tachometer.

3.15.1 Exhaust. The engine shall be equipped with exhaust muffler and stack. Exhaust stack outlets shall be provided with rain caps, 45° or 90° bends. Maximum sound level shall not exceed the recommended level of SAE J952.

3.16 Transmission and torque converter. The unit shall have a full torque power shift transmission. The transmission system shall provide not less than eight forward and two reverse speeds. Torque converter oil coolers shall be furnished. Coolers shall maintain the oil entering the converters at a temperature not exceeding 240°F. The unit shall be furnished complete with the following accessories:

a. Full flow oil filter(s).
b. Torque converter oil temperature indicator.

3.17 Axles. The unit shall be a 4 by 2 combination with a single engine on the tractor with front axle drive coupled to tow the trailer with a nondriven axle. The axle of the tractor shall be equipped with either a no-spin differential, or automatic torque proportioning differential. Planetary positive reduction gearing or direct spur gear reduction gearing shall be utilized for all drive axles.

3.18 Steering mechanism. Steering mechanism for the unit shall be full power steering. The steering mechanism shall be capable of turning the tractor 90° right or left of the longitudinal centerline of the trailer. An emergency steering system conforming to SAE J53 shall be furnished to provide continued steering in the event of failure of steering power source.

3.19 Hydraulic system. A hydraulic system shall be utilized for transmitting engine power to the mechanisms that operate the trailer dump cylinders and tractor steering. The hydraulic system shall be complete with all operating accessories, including pump(s), reservoir(s), valves, filters, hoses, piping, fittings, and pressure relief valve(s). Pump(s) shall supply a continuous flow of hydraulic fluid, and shall have sufficient capacity to energize all hydraulically powered components. The hydraulic system shall be capable of withstanding a bursting pressure of four times the working pressure. The system, including hoses, shall be suitable for operating temperatures ranging from -20°F to +250°F. Hoses shall be protected against chafing through utilization of supports and protective devices. Hydraulic system shall be protected against the entrance of dirt or other foreign material. A wiper shall be provided on each hydraulic cylinder located near the piston rod seal to prevent dirt and abrasives from damaging the seal or entering the cylinder. System shall be furnished with full flow filter(s) having a micron rating not greater than 25. When an open type hydraulic system is furnished, reservoir(s) shall be vented by a separate vent or through the filler cap. In either case, the vent shall be equipped with a micronic filter element to prevent dirt from entering the reservoir. Filter shall have a micronic rating of not greater than 25. Reservoir(s) shall have sufficient capacity to allow for maximum displacement of all cylinders at the same time, and to aid heat dissipation. Return line shall be terminated below the minimum oil level of the reservoir. Suction line shall be above the reservoir bottom. On open

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.

type hydraulic systems, the suction line shall be protected with a readily accessible strainer. Dump control valve shall be four-position and provide for power raise, lower, hold, and float.

3.20 Operator's station. Operator's station shall consist essentially of a shock-absorbing operator seat, steering wheel, a safety glass windshield equipped with power operated windshield wiper(s), instrument panel, operating controls, and a rollover protective structure conforming to SAE J1040. The seat and back rest shall be covered with a durable waterproof covering. The seat shall be equipped with a seat belt conforming to SAE J386. All controls and instruments that are not already identified to show their intended function of operation shall be clearly identified for their intended operation and function with permanently affixed identification plates of noncorrosive material. Decals shall not be used. Controls shall be located within the dimensional spaces specified in SAE J898. Instruments shall be readily visible to the operator when seated in a normal operating position. Instruments shall be protected against damage by an anti-vandalism panel guard, unless the unit is equipped with an enclosed cab with lockable door.

3.20.1 Cab. When specified (see 6.2), a fully enclosed cab of all steel or fiberglass construction shall be furnished. The cab shall have a hinged door and shall enclose the operator's seat and all controls, and shall not obstruct the view of the operator while seated. Closed cab accessories shall include a rearview mirror of the adjustable truck type, with minimum size of 4 by 8 inches, mounted on the left exterior of the cab. The enclosed cab shall be furnished with a heater and defroster. The heater and defroster shall be the manufacturer's standard for maintaining the operator's comfort in temperatures down to -20°F.

3.21 Lighting. The lighting system shall be either 12 or 24 volt, but shall be compatible with the voltage of battery(s) and alternator including the following:

a. Two headlamps (minimum).
b. Instrument panel lamps to illuminate all instruments.
c. One rearward directed floodlamp.
d. Two taillights and two stoplights. These lights may be a combination of both.

3.22 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.23 Toolbox. The unit shall be provided with a toolbox made of sheet steel. The toolbox shall have a hinged lid which shall open not less than 90°, and a drawbolt or spring loaded latch of a type that will keep the lid closed when the toolbox is subjected to vibration. Toolbox shall be securely fastened to the machine in a protected and accessible location. The toolbox shall be of sufficient size to hold the tools normally used to maintain the unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.

3.24 Lifting and tying down attachments. When specified (see 6.2), unit shall be equipped with lifting and tying down attachments. Attachments shall conform to MIL-STD-209, type III. Tying down attachments may be identified by stenciling or other suitable marking of the unit. Tying down marking shall clearly indicate the attachments are intended for tying down of the unit on the carrier when shipped. Attachments shall be indicated by a transportation plate. Transportation plate shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the unit showing the center of gravity shall be provided on the transportation plate.

3.25 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated as specified in section 5. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 pounds per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.26 Instruction plates. The unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the unit.

3.27 Identification plate. An identification plate will be furnished by the contracting officer for each unit. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each unit in a conspicuous place with corrosion-resistant screws, rivets, or bolts not less that 1/8 inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.28 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be green No. 14064, conforming to FED STD No. 595. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied on a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.29 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.

3.30 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.31 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be an approved shape and shall be concentric with the rivet. Rivet head shall makefull contact with the surface to be joined. The rivetrs shall be so fabricated as to develop a joint strength not less than the design value.

3.32 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.33 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.34 Workmanship. The quality of workmanship shall be such as to produce units that are in accordance with the requirements of this specification and that are so constructed as to insure proper functioning of all parts of the unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete unit when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.3 and the applicable

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.

tests of 4.4, and, when specified (see 6.2), the preproduction pack inspection of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the test of 4.4.2, and the packaging inspection of 4.5.

4.3 Examination. Each unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The first article unit shall be tested as specified in 4.4.1, and each production unit shall be subjected to the operational test specified in 4.4.2.

4.4.1 First article tests. The first article unit shall be subjected to the tests specified in 4.4.1.1 through 4.4.1.7. Failure to pass any phase of the test shall be cause for rejection.

4.4.1.1 Performance tests. Tests shall, as a minimum, assure that the unit is capable of meeting the performance requirements as specified in section 3. The unit shall be operated for not less than 8 hours. The unit shall be loaded to a capacity of not less than 66,000 lb per load and each load shall be hauled and dumped, without assistance, from auxiliary equipment. All hauling and traveling shall be at maximum safe speed.

4.4.1.2 Load-haul-dump test. Test shall consist of approximately 50 load-haul-dump cycles with a one-way haul distance of not less than 1,500 feet.

4.4.1.3 Gradeability test. The unit loaded to the specified struck capacity, shall negotiate an approximately 100-foot long, well compacted earth grade, having a minimum slope of 20 percent. The unit shall travel up and down this slope at least three times and come to a complete stop with the entire unit on the grade not less than two times going up and two times going down the 20 percent grade.

4.4.1.4 Side hill slope test. Without payload, the unit shall traverse longitudinally along a grade having a minimum slope not less than 30 percent. The grade shall be of compacted earth, having a minimum length not less than 100 feet. Unit shall make not less than two passes; one from right to left and one from left to right.

4.4.1.5 Travel speed test. The unit shall have an empty travel speed not less than 24 mph on a dry, level, earthen haul road. After completion of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.

test, the entire unit shall be examined for evidence of excessive heating, failure, slippage, malfunction, or deformation. Failure to pass any phase of the examination and testing shall be cause for rejection.

4.4.1.6 Brake tests. The service brakes and parking brake shall be tested to determine conformance to 3.13.1 and 3.13.2.

4.4.1.7 Lifting and tying down attachment test. When furnished, lifting and tying down attachments shall be tested to determine conformance to 3.23. The manufacturer may furnish certified engineering calculations in lieu of a test to provide that the lifting and tying down attachments offered meets the requirements set forth in MIL-STD-209.

4.4.2 Operational test. Each unit shall be completely assembled, properly adjusted, and otherwise serviced for operation. All controls shall be operated as many times as necessary to determine ease of operation, effectiveness, and responsiveness, and that the mechanisms actuated by the controls operate promptly, fully, and without restriction, malfunction, or vibration. Gearcases, bearings, and other moving parts shall be tactually examined for heating or abnormal operation insofar as is practicable and possible, Entire unit shall be examined for evidence of failure, slippage, or deformation. Failure to pass any phase of this test shall be cause for rejection.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-T-3351. The inspection shall consist of the quality conformance inspection, and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-T-3351 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. The unit covered by this specification is intended for earthmoving operations in construction projects.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
c. When ether priming system is required (see 3.10).
d. When enclosed cab is required (see 3.20.1).
e. When fungus resistance is required (see 3.22).
f. When lifting and tying down attachments are required (see 3.24).
g. When a preproduction pack inspection is required and the time frame required for submission (see 4.2.1 and 4.5).
h. Level of preservation and level of packing required (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Definitions.

6.5.1 Maintained earth haul road. A maintained earth haul road has a smooth, firm, level, earth surface that undulates or slightly flexes under load, and is watered and maintained regularly.

6.5.2 Gross vehicle weight. The gvw shall consist of the weight of the complete unit with a full complement of fuel, lubricants, and coolant, a 175-lb operator, and the minimum payload as specified in table I.

Preparing activity:

Navy - YD

Project No. 3805-N110

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer (156)
Naval Construction Battalion Center
Port Hueneme, CA 93043

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. 3805-N110

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-T-29220(YD) - TRAILER, SEMITRACTOR, END DUMP, 22 CUBIC YARD CAPACITY, DED

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 Replaces edition of 1 Jan 72 which may be used. S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:20Z
Check the source to verify that this is the current version before use.